Numerical Bosonic String Integrands from One-Face Ribbon Graphs

GPT 5.5
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April 29, 2026

Abstract

This note extracts the numerical construction of bosonic string integrands from the local string Monte Carlo note, the Markdown implementation notes, and the Python scripts in the covariant ribbon-graph repository. The purpose is not to replace the draft, but to isolate the precise mathematical object computed by the code, the setup in which it represents a continuum worldsheet CFT quantity, the one regulator assumption, and the checks that support the construction.

The central object is a Riemann surface obtained by sewing the boundary of a disk according to a one-face metric ribbon graph. The matter determinant and compact-boson winding sum are computed directly from a discretized identity wavefunctional on the disk. The holomorphic data, period matrix, prime form, Riemann constants, and bc correlators are reconstructed from the same disk sewing. These data give a numerical representative of the matter and ghost factors in the bosonic string measure, up to the standard issues of Weyl-frame renormalization, moduli-independent normalization, automorphism factors, and the treatment of ribbon-cell boundaries. The native moduli space in this note is $\mathcal{M}_{g,1}$; for partition-function applications the marked puncture carries the identity operator.

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1 Scope and status

The sources reviewed for this note are:

- the local string Monte Carlo note `covariantformalism/tex/stringMC.tex`;
- the Strebel/ribbon working note `Strebel.tex` and the draft outline in `draft/`;
- the Markdown implementation notes `covariantformalism/markdown/riemann_surface_tools.md` and `covariantformalism/markdown/genus3_bc_chi18_ratio_check.md`;
- the numerical scripts in `covariantformalism/python/`, especially the graph-generation, determinant, compact-boson, period-matrix, Riemann-surface, scaling-sweep, and Mumford-check scripts listed explicitly in Section 11;
- the relevant sections of the external string notes source `https://github.com/xiyin137/stringbook/`, namely conformal gauge with moduli, the holomorphic reformulation of b-ghost insertions, the ghost number anomaly, and the higher-loop bosonic string measure.

I will use the following status labels.

Exact CFT. A statement follows from the continuum CFT or string measure formalism, independent of the lattice regulator.

Setup. A statement fixes the class of graphs, coordinates, or normalization conventions used in this note.

Theorem/geometric input. A statement is a standard mathematical input, for example the Strebel-Kontsevich cell decomposition or a continuum CFT identity.

Regulator assumption. A statement asserts convergence of the discretized disk construction to the continuum quantity after allowed local counterterms. This is the only nontrivial assumption made by the numerical construction.

Check. A statement records a diagnostic already implemented in the repository.

In the body below these labels appear either as theorem-like environments or as bold lead-ins to the relevant paragraphs. They are a bookkeeping device for separating definitions from continuum facts, numerical evidence, and the single regulator assumption.

The output of the present construction should be read as a local integrand or section over the moduli represented by a fixed-perimeter one-face ribbon cell, namely a cell in $\mathcal{M}_{g,1}$. It is not, by itself, a completed global integral until the cell measure, automorphism factors, and boundary strata are treated consistently.

2 Setup, theorems, and the regulator assumption

Setup. Unless stated otherwise, the numerical formulas use $\alpha' = 1$. Factors of α' are restored in a few continuum formulas to identify the convention. The worldsheet is a closed oriented genus g Riemann surface Σ. The point of this section is to separate definitions and standard theorems from the one real analytic assumption in the numerical work.

Definition 1 (Ribbon-cell setup)**.** *The primary numerical coordinate system is a positive-length, one-face metric ribbon graph Γ of genus g. On the open top-dimensional cells the graph is trivalent, and hence*

$$E = 6g - 3, \qquad V = 4g - 2, \qquad F = 1. \tag{2.1}$$

Degenerate cells with vanishing edge lengths are boundary strata and are not part of the present local construction.

Theorem 1 (Strebel-Kontsevich input)**.** *For a genus g Riemann surface with one marked point and one positive perimeter, the Strebel differential determines a one-face metric ribbon graph; conversely the corresponding metric ribbon graphs give the standard cell decomposition of*

$$\mathcal{M}_{g,1} \times \mathbb{R}_+, \tag{2.2}$$

with the usual orbifold identifications by graph automorphisms.

Remark 1 (Marked point and perimeter)**.** *The preceding theorem is not an assumption of the numerical method; it is the geometric setup. In this note we take $\mathcal{M}_{g,1}$, with fixed Strebel perimeter, as the native moduli space. For partition-function applications the marked puncture is assigned the identity operator. Thus the marked point is part of the data being integrated over, rather than something to be silently removed. If one later wants an unmarked vacuum amplitude on $\mathcal{M}_g$, one must specify a separate pushforward or fiber-normalization convention.*

Definition 2 (Weyl frame, zero modes, and winding sector)**.** *The lattice regulator is defined in the disk frame associated with the Strebel differential. Individual matter or ghost partition functions are therefore Weyl-frame dependent, while Weyl-invariant combinations, ratios at fixed central charge, and the critical matter-plus-ghost combination are the intended continuum quantities.*

For a noncompact boson, the constant zero mode is factored as a target-space volume and the determinant is taken on the orthogonal complement of constants. For a compact boson $X \sim X + 2\pi R$, the constant zero mode contributes $2\pi R$. Winding around the marked puncture is set to zero; equivalently, only edge-shift configurations satisfying the vertex incidence constraints are summed. These are conventions defining the regulated quantity.

Definition 3 (Holomorphic data conventions)**.** *Period matrices are computed from A-normalized holomorphic one-forms. Abel maps use a fixed basepoint in the disk, and physical formulas only use*

basepoint-independent combinations. Prime forms use an odd theta characteristic and principal square-root branches; the independence of the prime form from the odd characteristic is a standard theorem and is checked numerically.

For $g \geq 2$, the note and the current checks use the canonical construction of the Riemann constant vector described in the implementation notes: zeros of an A-normalized holomorphic one-form determine a canonical divisor, and half-period candidates are selected by Riemann-vanishing tests. The higher-genus σ normalization is fixed from the $\lambda = 1$ Verlinde-Verlinde identity together with the chiral determinant convention. The older direct cycle-integral routine is not used as an input to the claims in this note.

Definition 4 (Normalization convention)**.** *All absolute amplitudes in this note are defined only up to constants that are independent of moduli and of the compactification radius. The implemented checks compare ratios or local sections for which these constants cancel.*

Regulator Assumption 1 (Continuum limit modulo local counterterms)**.** *For integer edge lengths, set*

$$m = \sum_{e=1}^{E} \ell_e, \qquad L = 2m. \tag{2.3}$$

The continuum limit is taken by scaling all edge lengths by a common large integer while keeping their ratios fixed. The nontrivial analytic assumption is that the resulting discretized disk path integral converges to the intended continuum CFT quantity after multiplication by local counterterms allowed by the symmetries of the regulator, such as a lattice cosmological term proportional to L and an Euler-curvature term. The remaining freedom is a finite moduli-independent normalization.

The large-L fits used in the code are the practical implementation of this assumption. They require all edge lengths to be large enough that short-edge artifacts are negligible; the implementation notes use minimum edge lengths of order 200 for stable fits.

Remark 2. *There is a counting-convention issue in the working files. The draft outline cites the Bacher–Vdovina counts 9 and 1726 for one-vertex genus-two and genus-three triangulations. The current compact-boson file* `compact_partition.py` *stores 9 genus-two one-face graphs, and the current genus-three data file* `genus3_ribbon_graph_data.txt` *declares 1726 stored one-face genus-three ribbon graphs. The current generator describes its output as isomorphism-reduced under graph automorphisms. An older sentence in* `stringMC.tex` *says that the compact-boson T-duality check was run over 4 genus-two topologies and 818 genus-three topologies. The reviewed repo files do not identify the enumeration convention behind the older 4 and 818 numbers. Until this is audited, the current-code counts should be quoted as 9 and 1726, and the older 4 and 818 statements should be treated as unresolved. No formula below depends on these counts.*

3 Continuum string measure

Exact CFT. This section records the continuum target of the numerical construction. In conformal gauge, after fixing puncture positions and a fiducial family of metrics $\hat{g}(t)$, the Faddeev-Popov procedure gives the usual insertion

$$\mathcal{B}_{t^k} = \frac{1}{4\pi} \int_\Sigma \mathrm{d}^2\sigma \, \sqrt{\hat{g}} \, b^{ab} \frac{\partial \hat{g}_{ab}}{\partial t^k}. \tag{3.1}$$

Thus a genus h, n-point bosonic string amplitude has the schematic form

$$\mathcal{A}_h[V_1,\dots,V_n] = N_{h,n}\int_{\mathcal{M}_{h,n}} \Omega_{6h-6+2n}, \qquad \Omega = \left\langle \mathrm{e}^{\mathcal{B}} \prod_i \mathcal{V}_i \right\rangle_\Sigma , \tag{3.2}$$

where

$$\mathcal{B} = \sum_k \mathrm{d}t^k\, \mathcal{B}_{t^k}. \tag{3.3}$$

This is the form used in the external string notes.

The holomorphic reformulation is essential for the numerical approach. If the complex structure is described by transition functions $z_i = f_{ij}(z_j;t)$, then (3.1) can be written only in terms of those transition functions:

$$\mathcal{B}_{t^k} = \sum_{(ij)} \int_{C_{ij}} \left[\frac{\mathrm{d}z_i}{2\pi\mathrm{i}}\, b_{z_i z_i}\, \frac{\partial z_i}{\partial t^k}\bigg|_{z_j} - \frac{\mathrm{d}\bar{z}_i}{2\pi\mathrm{i}}\, b_{\bar{z}_i\bar{z}_i}\, \frac{\partial \bar{z}_i}{\partial t^k}\bigg|_{\bar{z}_j} \right]. \tag{3.4}$$

This formula is the continuum reason that a construction based on sewing maps and holomorphic data is enough to recover the string integrand.

For a genus $g \geq 2$ vacuum amplitude in complex coordinates τ^a, $a = 1,\dots,3g-3$, the top form is

$$\Omega_{6g-6} = \left\langle \prod_{a=1}^{3g-3} \mathrm{d}\tau^a\, \mathrm{d}\bar{\tau}^a\, \mathcal{B}_{\bar{\tau}^a}\mathcal{B}_{\tau^a} \right\rangle_\Sigma^{\text{matter+ghost}} . \tag{3.5}$$

This unmarked formula is the standard continuum benchmark. The one-face ribbon construction used below works instead on $\mathcal{M}_{g,1}$. For the partition-function application we regard the marked puncture as carrying the identity operator, so the corresponding top form has degree

$$\dim_{\mathbb{R}} \mathcal{M}_{g,1} = 6g-4. \tag{3.6}$$

Equivalently, in local complex coordinates u^A, $A = 1,\dots,3g-2$, on $\mathcal{M}_{g,1}$, one may write schematically

$$\Omega^{\mathbf{1}}_{6g-4} = \left\langle \prod_{A=1}^{3g-2} \mathrm{d}u^A\, \mathrm{d}\bar{u}^A\, \mathcal{B}_{\bar{u}^A}\mathcal{B}_{u^A}\, (c\widetilde{c}\,\mathbf{1})(p) \right\rangle_{\Sigma,p}^{\text{matter+ghost}} , \tag{3.7}$$

where p is the marked point. In a coordinate patch split as (τ^a, p), the two B-insertions associated with moving p strip the $c\widetilde{c}$ factor and leave an integrated identity insertion. This is a marked partition-function density, not a claim that the identity insertion is an on-shell scattering vertex. More explicitly, the puncture-moving Beltrami differentials reduce to the local contour modes b_{-1} and $\widetilde{b}_{-1}$ around p; their action on the $c\widetilde{c}$ at the marked point supplies the local Jacobian for integrating over the marked fiber. This is the usual BRST coordinate-split descent from an unintegrated marked insertion to an integrated insertion. If period-matrix coordinates are available, the b-ghost zero modes are paired with holomorphic quadratic differentials. In genus two the quadratic differentials may be written

$$S_1 = \omega_1^2, \qquad S_2 = \omega_2^2, \qquad S_3 = \omega_1\omega_2, \tag{3.8}$$

with A-normalized holomorphic one-forms ω_I. Writing V_X for the noncompact target-space zero-mode volume, the genus-two critical bosonic string vacuum form is

$$\Omega_6 = -\frac{V_X}{(4\pi^2\alpha')^{26}(\det \operatorname{Im}\Omega)^{13}} \left| \frac{2\pi\mathrm{i}}{\chi_{10}(\Omega)} \mathrm{d}\Omega_{11}\mathrm{d}\Omega_{22}\mathrm{d}\Omega_{12} \right|^2 , \tag{3.9}$$

up to the standard overall normalization convention. Equation (3.9) is the principal genus-two analytic benchmark for the ghost and determinant machinery.

4 One-face Strebel cells and disk sewing

By the Strebel theorem, a point of $\mathcal{M}_{g,1} \times \mathbb{R}_+$ can be represented by a metric ribbon graph with one face. In the generic top-dimensional cell the graph is trivalent. Let the oriented boundary of the unique face be traced once. Each edge e appears twice in this boundary word, with opposite orientations. The disk model places the boundary word on $|z| = 1$, with integer edge length ℓ_e, so that the boundary has $L = 2\sum_e \ell_e$ lattice sites.

Let S_a, $a = 1, \ldots, 2E$, be the oriented boundary segments. If the two occurrences of edge e are $S_{a(e)}$ and $S_{b(e)}$, sewing imposes an orientation-reversing identification between the corresponding arcs of the unit circle. In the noncompact boson this is simply equality of the boundary field values. In the compact boson it is equality up to an integral shift, which will be described in Section 6.

The boundary prevertices are the points of $|z| = 1$ where adjacent boundary segments meet. There are

$$N_{\mathrm{pv}} = 2E = 3V = 12g - 6$$

such points in a trivalent one-face cell; triples of boundary prevertices are identified to the V trivalent surface vertices after sewing. If q_r, $r = 1, \ldots, N_{\mathrm{pv}}$, denote the boundary prevertices, a holomorphic one-form on the sewn surface pulls back to the disk with cubic-root singularities at these points. The numerical ansatz used by the improved higher-genus solver has the form

$$\omega(z) = \left[\prod_{r=1}^{N_{\mathrm{pv}}} \left(1 - \frac{z}{q_r}\right)^{-1/3}\right] P(z)\,\mathrm{d}z, \tag{4.1}$$

where P is represented by a polynomial in the truncated disk basis. The branch choices are those inherited from evaluating the singular factor inside the disk. The exponent $-1/3$ has a local geometric origin. At a trivalent surface vertex, the three boundary prevertices in its equivalence class contribute three disk half-neighborhoods, with total angle 3π before passing to a smooth surface coordinate. On each branch, a smooth local coordinate ξ is therefore related to the disk coordinate by $\xi \sim (z-q_r)^{2/3}$, and a regular holomorphic differential $\mathrm{d}\xi$ pulls back as $(z-q_r)^{-1/3}\mathrm{d}z$ times a regular factor.

Remark 3. *The lattice data are a regulator for the CFT on the sewn Riemann surface, not a piecewise-linear approximation to the final worldsheet metric. The continuum complex structure is obtained from the sewing, while the Strebel/disk metric only fixes a Weyl frame in which the regulator is defined.*

5 Matter determinant from the identity wavefunctional

For one free noncompact boson on the disk, the boundary field

$$X(\sigma) = x_0 + \sum_{n\neq 0} X_n \mathrm{e}^{\mathrm{i}n\sigma}$$

has identity wavefunctional

$$\Psi_1[X] \propto \exp\left[-\frac{1}{\alpha'}\sum_{n\geq 1} nX_nX_{-n}\right]. \tag{5.1}$$

The scripts use a circulant lattice kernel for this quadratic form. Up to an overall convention-dependent factor, the kernel is

$$\widehat{K}_L(r) = \frac{\sin(\pi/L)}{L\left[\cos(2\pi r/L) - \cos(\pi/L)\right]}, \qquad r \in \mathbb{Z}/L\mathbb{Z}. \tag{5.2}$$

Let K_L denote the associated circulant matrix,

$$(K_L)_{ij} = \widehat{K}_L((j-i) \bmod L), \tag{5.3}$$

with indices understood modulo L. This matrix has the constant vector as its null vector and approaches the continuum Dirichlet-to-Neumann quadratic form in the low Fourier modes. The files `partition_function.py` and `compact_partition.py` differ by a factor of 2 in this raw matrix; this is a regulator normalization convention, provided it is used consistently when comparing determinants and winding forms. With the sign convention in (5.2), the formula agrees with `direct_mat_n_fast` in `partition_function.py`; the corresponding routine in `compact_partition.py` returns one half of the same matrix. The positive object used in determinant computations is the sewn, reduced quadratic form $A'_{\Gamma,L}$, after removal of the constant zero mode.

Let $X \in \mathbb{R}^L$ be the vector of boundary values before sewing. Sewing the disk boundary is represented by a linear map

$$X = M_\Gamma y$$

from independent real variables y to the full boundary vector. The quadratic form on the sewn surface is

$$A_{\Gamma,L} = M_\Gamma^T K_L M_\Gamma. \tag{5.4}$$

It still has one null vector, the constant mode. If Q is any matrix whose columns span a complement to constants, define

$$A'_{\Gamma,L} = Q^T A_{\Gamma,L} Q. \tag{5.5}$$

Then the regulated noncompact one-boson partition function has the form

$$Z^{\mathrm{nc}}_{\Gamma,L} = C_L\, V_X\, (\det A'_{\Gamma,L})^{-1/2}, \tag{5.6}$$

where C_L is a local regulator constant and V_X is the target-space zero-mode volume. For D noncompact bosons the determinant factor is raised to the D-th power.

6 Compact boson and the winding lattice

Let $X \sim X + 2\pi R$. For each edge e, the two sewn boundary segments may differ by a shift $2\pi R s_e$ with $s_e \in \mathbb{Z}$. Around every cubic vertex the shifts must close. With an orientation on the graph, this closure condition is the integer incidence equation

$$Ds = 0, \qquad s \in \mathbb{Z}^E, \tag{6.1}$$

where D is the $V \times E$ oriented incidence matrix. Since the graph is connected,

$$\operatorname{rank} D = V - 1, \qquad \dim_{\mathbb{Z}} \operatorname{Ker} D = E - V + 1 = 2g.$$

Choose an integral basis

$$B : \mathbb{Z}^{2g} \longrightarrow \operatorname{Ker}_{\mathbb{Z}} D, \qquad s = Bn. \tag{6.2}$$

In the code this basis is constructed from graph cycles, equivalently from a spanning tree or from the stored graph incidence data.

In a fixed shift sector, the boundary field can be written

$$X = M_\Gamma y + 2\pi R\, W s, \tag{6.3}$$

where W inserts the chosen shifts along the identified sites. Substitution into the disk quadratic form gives

$$S_L(y, s) = \frac{1}{2} y^T A' y + 2\pi R\, y^T U s + (2\pi R)^2 \frac{1}{2} s^T C s. \tag{6.4}$$

After integrating over y on the nonzero-mode subspace, the winding sector is weighted by the Schur complement

$$T_{\text{Schur}} = C - U^T (A')^{-1} U, \qquad T'_{\text{Schur}} = B^T T_{\text{Schur}} B. \tag{6.5}$$

The compact-boson scripts use a normalized matrix, denoted T' below and stored as `T_reduced`, for which all factors from the $2\pi R$ shifts, the factor $1/2$ in the Gaussian action, and the raw-kernel normalization have already been absorbed. Equivalently, T' is defined by the winding weight

$$\exp[-4\pi R^2 n^T T' n].$$

With this convention,

$$Z^{(g)}_{\Gamma,L}(R) = 2\pi R\, C_L (\det A')^{-1/2} \Theta_{2g}(4\mathrm{i} R^2 T'), \tag{6.6}$$

where

$$\Theta_N(\Xi) = \sum_{n \in \mathbb{Z}^N} \exp\left(\pi \mathrm{i}\, n^T \Xi n\right). \tag{6.7}$$

For $\Xi = 4\mathrm{i} R^2 T'$, the summand is $\exp[-4\pi R^2 n^T T' n]$.

The continuum compact-boson partition function on a genus g surface with period matrix Ω, in the same $\alpha' = 1$ convention and up to a radius-independent factor, is

$$Z^{(g)}_{\text{comp}}(\Omega, R) = 2\pi R\, Z^{(g)}_{\text{bos}}(\Omega) \sum_{m,n \in \mathbb{Z}^g} \exp\left[-\pi R^2 (m + \Omega n)^T (\operatorname{Im}\Omega)^{-1} (m + \bar{\Omega} n)\right]. \tag{6.8}$$

The code checks that ratios of the lattice theta factor agree with (6.8). For a general positive matrix, Poisson summation does not return the same theta series; it returns the theta series of the dual quadratic form. Let $N = 2g$ and $G = 4T'$. Then

$$\Theta_N(\mathrm{i} R^2 G) = R^{-N} (\det G)^{-1/2} \sum_{m \in \mathbb{Z}^N} \exp\left[-\pi R^{-2} m^T G^{-1} m\right]. \tag{6.9}$$

Thus a same-matrix T-duality statement is equivalent to a self-duality condition on the normalized integral winding lattice: G^{-1} must be integrally equivalent to G, i.e. $U^T G^{-1} U = G$ for some $U \in GL(N, \mathbb{Z})$. This includes $\det G = 1$. More generally, any residual factor would have to be independent of both R and the moduli to be absorbed into the overall convention. The matrix

T' stored as `T_reduced` is normalized so that the following identity is the numerical diagnostic being tested:

$$R^g \Theta_{2g}(4\mathrm{i}R^2 T') = R^{-g}\Theta_{2g}(4\mathrm{i}R^{-2}T'), \tag{6.10}$$

which implies

$$Z^{(g)}(R) = R^{2-2g} Z^{(g)}(R^{-1}) \tag{6.11}$$

with the compact zero-mode factor included. In this note (6.10) is used as a numerical check of the compact-boson construction and of the normalization of the winding lattice, not as an independent theorem about arbitrary positive matrices T'.

7 Holomorphic data from the sewn disk

The same ribbon data determine the period matrix and the ingredients of the Verlinde-Verlinde bc correlator.

7.1 Holomorphic one-forms

The improved higher-genus solver in `ell_to_tau.py` uses (4.1). Let $m = \sum_e \ell_e$. The regular part is expanded as

$$P_I(z) = \sum_{a=0}^{m-1} c_{I,a} z^a. \tag{7.1}$$

For each edge, the two boundary arcs are sampled at half-integer lattice points. The pulled-back differential must match under orientation-reversing sewing. In the implemented convention the seam equation has the schematic form

$$\omega(z_R) + \omega(z_L) = 0 \tag{7.2}$$

at the paired sample points, after including the singular factor. The code constructs the matrix of these homogeneous constraints, finds its g smallest singular directions, and then performs a Hermitian block solve to obtain a stable basis of g one-forms.

7.2 Homology and periods

For a one-face graph, each edge defines a chord connecting the two occurrences of that edge on the boundary word. Two such chords intersect exactly when their endpoints interleave along the boundary. This gives an antisymmetric integer matrix J on the free abelian group generated by edge chords. Its rank must be $2g$. The code searches for a symplectic basis

$$\{\alpha_I, \beta_I\}_{I=1}^{g}$$

made from raw edge chords or small integer combinations, satisfying

$$\alpha_I \cdot \alpha_J = 0, \qquad \beta_I \cdot \beta_J = 0, \qquad \alpha_I \cdot \beta_J = \delta_{IJ}. \tag{7.3}$$

Let ω_i, $i = 1, \ldots, g$, be the unnormalized forms found above. Using disk antiderivatives evaluated at edge midpoints, the code computes

$$A_{ij} = \oint_{\alpha_j} \omega_i, \qquad B_{ij} = \oint_{\beta_j} \omega_i. \tag{7.4}$$

The normalized period matrix is then

$$\Omega = A^{-1}B. \tag{7.5}$$

The normalized forms are $A^{-1}\omega$ in the same row convention. Numerical diagnostics check that Ω is symmetric and that $\operatorname{Im}\Omega$ is positive definite.

7.3 Abel map, prime form, and sigma

The Abel map is

$$\zeta_I(z) = \int_{z_0}^{z} \widehat{\omega}_I, \tag{7.6}$$

where z_0 is the fixed disk basepoint and $\widehat{\omega}_I$ are the A-normalized holomorphic one-forms. The Riemann theta function convention is

$$\theta[\epsilon,\delta](y|\Omega) = \sum_{n\in\mathbb{Z}^g} \exp\left[\pi\mathrm{i}(n+\epsilon)^T\Omega(n+\epsilon) + 2\pi\mathrm{i}(n+\epsilon)^T(y+\delta)\right]. \tag{7.7}$$

This is the convention implemented in the Python theta routine; other common half-characteristic conventions differ by characteristic-dependent phase factors. For an odd characteristic ν, the prime form is

$$E(z,w) = \frac{\theta[\nu](\zeta(z)-\zeta(w)|\Omega)}{\sqrt{\omega_\nu(z)\omega_\nu(w)}}. \tag{7.8}$$

The result is independent of the odd characteristic; this is tested numerically.

The Riemann constant vector Δ is constructed canonically for $g \geq 2$ using a canonical divisor from zeros of a normalized holomorphic one-form and a half-period search selected by Riemann vanishing. This replaces the older direct cycle-integral formula, which is known not to be reliable for $g \geq 2$.

The σ function entering the bc correlator is used through ratios and through a normalization fixed by the $\lambda = 1$ Verlinde-Verlinde identity. The auxiliary divisor points used for this normalization must be generic and disjoint from all b- and c-insertions. Same-surface ratios are checked to be stable under changing this auxiliary divisor. In practice the scripts reject choices in which divisor points collide with insertion points or make the relevant theta, prime-form, or sigma-normalization denominators numerically singular.

8 The bc system and the chiral determinant

For a $(b_\lambda, c_{1-\lambda})$ system, the selection rule is

$$n_c - n_b = (1-2\lambda)(g-1). \tag{8.1}$$

In the conventions used by the current implementation, Z_{ch} is the primary chiral determinant factor. The historical symbol Z_1 is then defined by

$$Z_1 := Z_{\mathrm{ch}}^{-2}, \qquad Z_{\mathrm{ch}} = Z_1^{-1/2}.$$

The holomorphic correlator is

$$\begin{aligned}
&\left\langle \prod_{i=1}^{n_b} b_\lambda(z_i) \prod_{j=1}^{n_c} c_{1-\lambda}(w_j) \right\rangle_\Sigma \\
&\quad = Z_{\mathrm{ch}}\, \theta\left(\sum_i \zeta(z_i) - \sum_j \zeta(w_j) - (2\lambda-1)\Delta \mid \Omega \right) \\
&\qquad \times \frac{\prod_{i<i'} E(z_i, z_{i'}) \prod_{j<j'} E(w_j, w_{j'}) \prod_i \sigma(z_i)^{2\lambda-1}}{\prod_{i,j} E(z_i, w_j) \prod_j \sigma(w_j)^{2\lambda-1}}.
\end{aligned} \tag{8.2}$$

The function `bc_correlator_geometric_factor` returns the theta, prime-form, and sigma product in (8.2); equivalently, it is the holomorphic correlator with the single $Z_{\mathrm{ch}} = Z_1^{-1/2}$ prefactor stripped off.

The chiral determinant is fixed from the renormalized matter determinant. If $\exp c_{\det}(\Omega)$ denotes the finite part extracted from $-\frac{1}{2}\log\det A'$, the canonical convention in the implementation notes is

$$|Z_{\mathrm{ch}}(\Omega)|^2 = [\det \operatorname{Im}\Omega]^{1/2}\, \mathrm{e}^{c_{\det}(\Omega)} \tag{8.3}$$

up to one moduli-independent normalization factor. Equivalently,

$$|Z_1|^2 = |Z_{\mathrm{ch}}|^{-4}.$$

This convention is important: older text may call Z_1 the chiral-boson partition function, while the current code uses $Z_{\mathrm{ch}} = Z_1^{-1/2}$ as the prefactor in (8.2).

9 Large-L renormalization

The raw lattice quantities are not finite as $L \to \infty$. The compact-boson working notes introduce the counterterm scheme by fitting the self-dual compact partition function,

$$\log Z^{(g)}_{\Gamma,L}(R=1) = c_{\mathrm{comp}}(\Omega) + \gamma L + \alpha(g)\log L + o(1). \tag{9.1}$$

The bc/Mumford pipeline uses a determinant-only fit:

$$-\frac{1}{2}\log\det A'_{\Gamma,L} = c_{\det}(\Omega) + \gamma_{\det} L + \alpha_{\det}\log L + o(1), \tag{9.2}$$

where the code fits $\gamma_{\det}$ and $\alpha_{\det}$ directly for the genus and graph families used in the ratio check. Thus $\mathrm{e}^{c_{\det}}$, not the compact theta sum, is the finite determinant factor used below in (8.3).

The empirical self-dual compact-partition values quoted in `Strebel.tex` and in the string Monte Carlo note are

$$\gamma \simeq \frac{1}{4}\log 2, \qquad \alpha_0 \simeq -0.86112, \qquad \alpha_1 \simeq -0.27778.$$

Here $\alpha(g) = \alpha_0 + \alpha_1 g$ refers to the compact self-dual fit (9.1). The actual Mumford checks do not assume the quoted α_0, α_1: they fit the determinant coefficient $\alpha_{\det}$ from A'-data in the selected genus. No analytic derivation of these constants was found in the local notes or scripts. Until a heat-kernel, Liouville, or regulator calculation is supplied, the decimal fit coefficients should be

treated as empirical inputs to this counterterm scheme. The script `genus1_to_4_scaling_sweep.py` is the current reproduction driver for the compact large-L fits; for the self-dual coefficients above one should run the sweep at $R = 1$ and record the chosen L-window, graph families, and theta cutoffs. With the current script defaults for the L-window, an explicit command is

```
./.venv/bin/python "covariant formalism/python/genus1_to_4_scaling_sweep.py" \
  --R 1.0 --L-start 2000 --L-stop 5000 --L-step 200 \
  --json-out /tmp/radius_1_scaling.json
```

The exact status of these constants is an open derivation problem for a final draft; they are not used below as an independent theorem. The corresponding compact-fit local counterterm convention is

$$S_{\mathrm{ct}} = \gamma L + (\alpha_0 + \alpha_1) \log L - \frac{\alpha_1}{8\pi} \log L \int_\Sigma \mathrm{d}^2\sigma \, \sqrt{g} \, R(g). \tag{9.3}$$

Since

$$\int_\Sigma \sqrt{g} \, R(g) = 4\pi(2 - 2g),$$

(9.3) reproduces $\alpha_0 + \alpha_1 g$ in (9.1). In the determinant-only scheme used for (8.3), the fitted finite determinant factor is

$$[(\det A')^{-1/2}]_{\mathrm{ren}} = \mathrm{e}^{c_{\mathrm{det}}(\Omega)}. \tag{9.4}$$

This is a regulator definition, not a proof of universal convergence. The evidence for the compact fit is that the same γ and genus-linear $\alpha(g)$ stabilize across graph families, and that the resulting finite part satisfies the genus-one Weyl-anomaly comparison with the analytic compact-boson partition function. The evidence for the determinant fit is the Mumford-form agreement after $\gamma_{\mathrm{det}}, \alpha_{\mathrm{det}}$ are fit from same-genus A'-families.

10 Numerical bosonic string integrand

The CFT part of the critical bosonic string measure can be represented in terms of a Mumford form. Let $q_a(z)$, $a = 1, \ldots, 3g-3$, be a basis of holomorphic quadratic differentials. For generic points z_b, define

$$D_q(z_1, \ldots, z_{3g-3}) = \det[q_a(z_b)]. \tag{10.1}$$

The numerical bc correlator and the renormalized determinant determine the unmarked local Mumford factor through

$$\left| \left\langle \prod_{b=1}^{3g-3} b(z_b) \right\rangle_\Sigma \right|^2 |Z_{\mathrm{ch}}(\Omega)|^{52} = C_g \, |D_q(z_1, \ldots, z_{3g-3})|^2 \, |\mu_g(\Omega)|^2. \tag{10.2}$$

Here C_g is moduli independent, and the bracket denotes the full ghost correlator in the convention of (8.2), including its single Z_{ch} prefactor. The additional factor $|Z_{\mathrm{ch}}|^{52}$ in (10.2) is the matter contribution of the 26 critical bosons. If one instead used the stripped geometric factor returned by `bc_correlator_geometric_factor`, the displayed power of $|Z_{\mathrm{ch}}|$ would have to be shifted accordingly. Equation (10.2) is the point-insertion version of the same section that appears after contracting b-ghost zero modes with Beltrami differentials in the string measure. In the coordinate-split description of the identity-marked $\mathcal{M}_{g,1}$ density, the unmarked Mumford factor above is the CFT scalar factor pulled back along the forgetful map to the underlying surface. The remaining marked coordinate is accounted for by the B-insertions and $c\widetilde{c}$ factor in (3.7).

For genus two,

$$\mu_2(\Omega) = \frac{2\pi \mathrm{i}}{\chi_{10}(\Omega)}. \tag{10.3}$$

The implemented ratio check is precisely

$$\frac{|\langle bbb\rangle_1|^2}{|\langle bbb\rangle_2|^2}\left(\frac{|Z_{\mathrm{ch},1}|^2}{|Z_{\mathrm{ch},2}|^2}\right)^{26} = \frac{|D_{q,1}|^2}{|D_{q,2}|^2}\frac{|\chi_{10}(\Omega_2)|^2}{|\chi_{10}(\Omega_1)|^2}, \tag{10.4}$$

up to numerical error.

For genus three, the six quadratic differentials are

$$\omega_1^2, \quad \omega_1\omega_2, \quad \omega_1\omega_3, \quad \omega_2^2, \quad \omega_2\omega_3, \quad \omega_3^2.$$

For nonhyperelliptic genus-three surfaces,

$$|\mu_3(\Omega)|^2 \propto \frac{1}{|\chi_{18}(\Omega)|}, \qquad \chi_{18} = \prod_{\delta \text{ even}} \theta[\delta](0|\Omega), \tag{10.5}$$

where the product is over the 36 even spin structures. The current genus-three check compares the two sides of (10.2) using (10.5); the documented relative errors are at the 10^{-4} to 10^{-3} level for the tested examples.

For genus four and above, the products $\omega_I\omega_J$ are not an independent basis without relations. On the generic nonhyperelliptic genus-four locus there is one quadric relation among the ten products. The script `genus4_bc_local_mumford_check.py` therefore performs a same-surface local-chart check. On a fixed surface, choose two generic nine-tuples of b-insertion points P_1 and P_2. For each usable chart obtained by omitting a quadratic monomial with nonzero quadric coefficient, form the two determinants $D_q^{(r)}(P_1)$ and $D_q^{(r)}(P_2)$. The diagnostic compares

$$\frac{|\langle b^9(P_1)\rangle|^2}{|\langle b^9(P_2)\rangle|^2} \quad \text{with} \quad \frac{|D_q^{(r)}(P_1)|^2}{|D_q^{(r)}(P_2)|^2}$$

on that same surface, and reports the relative error, the quadric residual, and the spread across usable omission charts. Hyperelliptic or otherwise special loci require separate treatment.

10.1 Compact radius dependence

For a $c = 1$ matter factor consisting of a compact boson of radius R and a radius-independent compensating CFT, the radius dependence of the fixed-surface integrand is

$$\mathcal{R}_R(\Sigma) = \frac{Z^{\mathrm{ren}}_{\mathrm{comp}}(\Sigma, R)}{Z^{\mathrm{ren}}_{\mathrm{comp}}(\Sigma, R_0)}. \tag{10.6}$$

At fixed surface, Weyl-frame and determinant factors cancel in this ratio except for the compact winding theta functions and the zero-mode factor. This is the cleanest numerical observable for testing the compact-boson part of the integrand. The self-dual radius $R_0 = 1$ is the most common reference in the local notes.

10.2 Cell-coordinate measure

Let ℓ_a, $a = 1, \ldots, E-1 = 6g-4$, be local real edge-length coordinates in a fixed-perimeter one-face ribbon cell. The pullback of the marked identity form (3.7) to the cell is schematically

$$\Phi_\Gamma^* \Omega_{6g-4}^{\mathbf{1}} = \mathcal{I}_\Gamma^{\mathbf{1}}(\ell)\, \mathrm{d}\ell_1 \cdots \mathrm{d}\ell_{6g-4}, \tag{10.7}$$

after imposing the fixed-perimeter condition. The scalar $\mathcal{I}_\Gamma^{\mathbf{1}}$ contains the matter determinant, compact winding sum if present, and the ghost/Mumford factor appropriate to the identity-marked surface. A global Monte Carlo integral over $\mathcal{M}_{g,1}$ must also include the cell automorphism factor $1/|\operatorname{Aut}\Gamma|$ and a consistent treatment of cell boundaries. The present ratio checks do not test the automorphism factor: they compare fixed-surface or same-cell scalar quantities, so $1/|\operatorname{Aut}\Gamma|$ cancels. The automorphism weight becomes a genuine input only in a global sum or integration over ribbon cells.

The current code has strong tests for the CFT scalar factors and local Mumford-section relations. A final production integrand should still make explicit the fixed-perimeter cell measure and the normalization convention for the identity-marked puncture. A later comparison with an unmarked integral on $\mathcal{M}_g$ would require an additional pushforward or fiber-normalization step, but that is not part of the native ribbon-cell prescription used here.

11 Algorithmic summary

For a fixed one-face ribbon graph Γ and integer edge lengths ℓ_e, the implemented pipeline is:

1. Trace the unique boundary word and build the oriented segment pairings.
2. Construct the matter sewing matrix $A'_{\Gamma,L}$ from the disk identity wavefunctional.
3. For a compact boson, build the incidence matrix D, an integral basis B of $\operatorname{Ker} D$, the Schur complement and script-normalized matrix T', and the theta function $\Theta_{2g}(4\mathrm{i}R^2T')$.
4. Reconstruct g holomorphic one-forms using the cubic-prevertex singular ansatz and the sewn-boundary matching equations.
5. Construct a symplectic homology basis from edge-chord intersections and compute the period matrix $\Omega = A^{-1}B$.
6. Build Abel maps, prime forms, canonical Riemann constants, and σ-values.
7. Evaluate the Verlinde-Verlinde bc correlator and insert the renormalized chiral factor Z_{ch}.
8. Combine the ghost correlator with the renormalized matter determinant to obtain the local Mumford/integrand factor, comparing to analytic forms when available.

File	Role
`Fast_Ribbon_Generator.py`	Enumeration of one-face ribbon-graph topologies, graph-isomorphism reduction, automorphism data, and genus-four topology generation used by the current diagnostics.
`partition_function.py`	Generic one-face sewing matrices, reduced matter determinant, and ghost wavefunctional experiments.
`compact_partition.py`	Compact-boson winding matrices, stored genus-one/genus-two graph data, and theta-function tests.
`ell_to_tau.py`	Improved holomorphic one-form solver, homology basis, period matrix, local prevertex factors, and genus-two modular forms.
`riemann_surface_tools.py`	Surface data object, Abel maps, prime form, canonical Riemann constants, σ, bc correlators, determinant fits, and chiral normalization.
`genus2_bc_igusa_ratio_check.py`	Genus-two check against the Igusa cusp form.
`genus3_bc_chi18_ratio_check.py`	Genus-three check against the χ_{18} relation.
`genus4_bc_local_mumford_check.py`	Genus-four same-surface local Mumford chart check using the quadric relation.

12 Checks and evidence

Check. The strongest existing checks are the following. Their accuracy is limited by different effects in different rows: finite L extrapolation and fit-window choice for determinant renormalization, theta-series truncation, numerical quadrature near singular prevertices, and conditioning of local determinant charts.

Source	Reproducibility data quoted in the notes
`Strebel.tex`, `stringMC.tex`, and `genus1_to_4_scaling_sweep.py`	At $R = 1$, the working compact partition-function fit targets $\gamma \simeq \frac{1}{4}\log 2$ and $\alpha(g) \simeq -0.86112 - 0.27778g$. No analytic derivation of these constants is contained in the local notes or scripts. The sweep script records the L-window, selected graph families, and theta cutoffs; run it with `--R1.0` for the self-dual scan.
`covariantformalism/markdown/riemann_surface_tools.md`	For Abel-jump consistency on stored genus-two topology 1, the reported worst componentwise errors are 6.14×10^{-6}, 1.63×10^{-6}, and 7.42×10^{-7} for edge lists $[50]^9$, $[100]^9$, and $[150]^9$, respectively.
`genus2_bc_igusa_ratio_check.py`	For $\ell_1 = [300]^9$ and $\ell_2 = (250, 250, 250, 250, 250, 250, 250, 250, 700)$, the documented Igusa comparison gives LHS $= 7.139760447638384$, RHS $= 7.139757771156295$, relative difference 3.75×10^{-7}.
`genus3_bc_chi18_ratio_check.py` and `covariantformalism/markdown/genus3_bc_chi18_ratio_check.md`	With topology 1, safe divisor points, and $n_{\max} = 4$, the documented χ_{18} comparisons against the symmetric modulus $M_1 = [300]^{15}$ give relative errors 4.91×10^{-4}, 5.96×10^{-4}, and 4.24×10^{-4} for the three listed comparison moduli.
`genus4_bc_local_mumford_check.py`	Reports a same-surface nine-b ratio, the local quadric-chart determinant ratio, the relative error, the quadric residual, and chart/divisor stability diagnostics. No representative numerical tolerance is fixed in this note; the claim supported here is local and generic, not a global genus-four integral.

Genus one compact boson. The lattice compact partition function agrees with the analytic torus lattice sum after mapping edge lengths to τ, and satisfies T-duality in the form (6.10).

Higher-genus compact boson. The incidence-kernel construction gives a rank-$2g$ winding lattice. The genus-g T-duality identity is then a numerical diagnostic: the notes report tests for all stored genus-two cases, the stored genus-three cases, and one genus-four topology.

Period matrices. The reconstructed Ω is symmetric up to numerical error and has positive definite imaginary part. Abel jumps across sewn boundary pairs land on the period lattice with errors decreasing as the edge lengths increase.

Prime form. Spin-structure independence of the prime form is checked numerically. The genus-two notes quote relative differences of order 10^{-6} or smaller in representative large-edge tests.

Genus two Mumford form. The Igusa-ratio check (10.4) agrees at the level of roughly 4×10^{-7} in the documented run.

Genus three Mumford form. The χ_{18} check uses the corrected squared-absolute relation

$$|\mu_3|^2 \propto \frac{1}{|\chi_{18}|}.$$

The documented runs agree at roughly 10^{-4} to 10^{-3}. The weaker precision compared with the genus-two Igusa check is consistent with the higher-dimensional theta sums, the sigma-normalization sensitivity, and the larger number of b-ghost insertions; it should be treated as a numerical accuracy budget to be reported, not as a new analytic assumption.

Genus four local chart. The same-surface nine-b ratio is implemented as a comparison against the determinant ratio in a local quadric chart, with chart spread used as a diagnostic of numerical stability. A draft-ready statement should quote a specific run, relative error, quadric residual, and divisor/chart stability tolerance.

13 Reference anchors

This note is primarily a code-to-continuum synthesis. A polished draft should replace the path-style references above by ordinary citations. The minimum source list should include the following anchors.

- Strebel's theory of quadratic differentials for the correspondence between metric ribbon graphs and punctured Riemann surfaces with perimeters.
- Kontsevich's matrix-Airy construction, and the Mulase–Penkava exposition, for the ribbon-cell decomposition and its relation to moduli-space forms.
- Faddeev–Popov, Friedan–Martinec–Shenker, Belavin–Knizhnik, and Polchinski for the covariant string measure, bc-ghost insertions, and the BRST descent between unintegrated and integrated vertex insertions.
- Mumford's determinant-line formalism, together with the Verlinde– Verlinde bosonization formulae, for the chiral measure and the ghost correlator used in the Mumford-form checks.
- Igusa's work on Siegel modular forms for the genus-two χ_{10} and genus-three χ_{18} modular forms used as analytic benchmarks.
- Deconinck–Patterson–Swierczewski for the canonical-divisor algorithm used in the current Riemann-constant computation.
- Bacher–Vdovina and the local ribbon-graph enumeration scripts for one-vertex triangulation counts and topology bookkeeping.

14 Open points for the draft

The following issues should be stated explicitly in any polished draft.

1. The one-face Strebel coordinates naturally describe $\mathcal{M}_{g,1} \times \mathbb{R}_+$. This note works natively on fixed-perimeter $\mathcal{M}_{g,1}$, with the marked puncture carrying the identity operator for partition-function applications. A comparison to an unmarked amplitude on $\mathcal{M}_g$ would require an additional pushforward or fiber-normalization convention.
2. The absolute normalization of the full amplitude is not fixed by the current determinant and ghost checks. Ratios and local Mumford sections are the controlled quantities.

3. The determinant finite part $e^{c_{\det}(\Omega)}$ is a renormalized quantity defined by the large-L fit. The fit should be treated as part of the regulator definition, with convergence tests reported for every production regime.

4. Numerical values quoted in the draft should be tied to a script, command line, graph or topology choice, edge-length data, theta cutoff, fit window, and code revision. The small source table in Section 12 is a starting point, not a replacement for a reproducibility table in a paper.

5. Lower-dimensional ribbon cells, automorphism factors, the fixed-perimeter cell measure, and boundary behavior of the integrand are required for a global Monte Carlo integral on $\mathcal{M}_{g,1}$.

6. The legacy one-form and Riemann-constant routines should not be used in the final pipeline. The improved cubic-singularity solver and canonical Riemann-constant construction are the versions supported by the current tests.

7. In genus four and higher, the quadratic-differential basis must account for canonical-curve relations on the relevant stratum. The genus-two and genus-three closed modular forms do not generalize naively.

8. The path-style references in this note should be converted into precise `\cite` entries in a polished draft, using the anchors in the preceding section as a starting point. When older notes disagree with current scripts or Markdown diagnostics, the current code and the checked numerical diagnostic should be treated as the source of record until the disagreement is resolved explicitly.